

MA 141: Calculus I

Fall 2026 · Second Session (10 weeks) · Mon Oct 5 – Fri Dec 11, 2026 · 4 credit hours

Instructor	Dr. Jordan Rivera
Email	jrivera@example.edu
Office	Mathematics Building 312
Office hours	Mondays and Wednesdays 2:00–3:30 PM, Fridays 9:00–10:00 AM, or by appointment
Class meetings	Mon / Wed / Fri 10:40–11:30 AM (lecture) and Tue 1:30–2:20 PM (recitation), Science Hall 1102
Teaching assistants	Priya Natarajan (pnatarajan@example.edu) — recitation and homework questions
Prerequisite	MA 107 or a placement score qualifying for calculus.

Course description

An introduction to differential calculus. Topics include limits and continuity, the derivative and its interpretations, differentiation rules, related rates, linear approximation, extreme values, the Mean Value Theorem, curve sketching, optimization and an introduction to antiderivatives. The course emphasizes both computation and understanding: you will explain why results are true, not just apply formulas.

Learning outcomes

By the end of this course you will be able to:

- Evaluate limits graphically, numerically and algebraically, and use them to determine continuity.
- Compute derivatives using the limit definition and the standard differentiation rules.
- Interpret the derivative as a rate of change and as the slope of a tangent line.
- Solve related-rates and optimization problems drawn from science, engineering and economics.
- Use the first and second derivative to analyze and sketch the graph of a function.
- Communicate mathematical reasoning clearly in writing.

Required materials

- Calculus: Early Transcendentals (OpenStax Calculus Volume 1, free online)
- A scientific calculator (graphing calculators are not allowed on tests)
- Access to the online homework system (link on the course website)

Grading

Component	Weight
Homework (8 sets, lowest dropped)	15%
Quizzes (4)	10%
Test 1	15%
Test 2	15%
Modeling project	15%
Final exam (cumulative)	25%
Recitation participation	5%
Total	100%

Grading scale: A 90–100 · B 80–89 · C 70–79 · D 60–69 · F below 60. Final grades are not curved; scores are rounded to the nearest whole percent.

Major project

Modeling project — “Calculus in the Real World.” In groups of three, choose a real situation (a rocket launch, a medicine dose, a business's profit) and model it with a function. Use derivatives to find rates of change and an optimal value, and explain what your answer means in context. Deliverables: a one-page proposal, a draft for feedback and a 4–6 page final report.

Course schedule

The schedule may change with notice; changes will be announced in class and on the course website.

Week	Dates	Topics	Due this week
1	Oct 5 – Oct 9	Functions review; the idea of a limit	Thu Oct 8: Homework 1 (Functions and limits)
2	Oct 12 – Oct 16	Limit laws, one-sided limits and continuity	Wed Oct 14: Quiz 1 (Limits) Thu Oct 15: Homework 2 (Limit laws and continuity)
3	Oct 19 – Oct 23	Squeeze Theorem, Intermediate Value Theorem, limits at infinity	Thu Oct 22: Homework 3 (Squeeze Theorem and IVT) Fri Oct 23: Project proposal due
4	Oct 26 – Oct 30	The derivative as a limit; Test 1	Thu Oct 29: Homework 4 (Definition of the derivative) Fri Oct 30: Test 1 (Limits and continuity)
5	Nov 2 – Nov 6	Power, product and quotient rules; derivatives of trig functions	Wed Nov 4: Quiz 2 (Differentiation rules) Thu Nov 5: Homework 5 (Product and quotient rules)
6	Nov 9 – Nov 13	Chain rule and implicit differentiation	Thu Nov 12: Homework 6 (Chain rule and implicit differentiation) Fri Nov 13: Project draft due
7	Nov 16 – Nov 20	Related rates and linear approximation	Wed Nov 18: Quiz 3 (Related rates) Thu Nov 19: Homework 7 (Related rates and linearization)

Week	Dates	Topics	Due this week
8	Nov 23 – Nov 27	Extreme values and the Mean Value Theorem; Test 2 (No class Wed Nov 25 – Fri Nov 27: Thanksgiving break)	Mon Nov 23: Test 2 (Derivatives and applications) Tue Nov 24: Homework 8 (Extrema and MVT)
9	Nov 30 – Dec 4	Curve sketching, optimization and L'Hôpital's Rule	Wed Dec 2: Quiz 4 (Optimization) Fri Dec 4: Final modeling project due
10	Dec 7 – Dec 11	Antiderivatives; review; final exam	Wed Dec 9: Final exam (cumulative)

Important dates and deadlines

Date	Item	Type	Time
Thu, Oct 8, 2026	Homework 1 (Functions and limits)	Assignment due	11:59 PM
Wed, Oct 14, 2026	Quiz 1 (Limits)	Quiz	in class
Thu, Oct 15, 2026	Homework 2 (Limit laws and continuity)	Assignment due	11:59 PM
Thu, Oct 22, 2026	Homework 3 (Squeeze Theorem and IVT)	Assignment due	11:59 PM
Fri, Oct 23, 2026	Project proposal due	Project milestone	11:59 PM
Thu, Oct 29, 2026	Homework 4 (Definition of the derivative)	Assignment due	11:59 PM
Fri, Oct 30, 2026	Test 1 (Limits and continuity)	Exam	in class
Wed, Nov 4, 2026	Quiz 2 (Differentiation rules)	Quiz	in class
Thu, Nov 5, 2026	Homework 5 (Product and quotient rules)	Assignment due	11:59 PM
Thu, Nov 12, 2026	Homework 6 (Chain rule and implicit differentiation)	Assignment due	11:59 PM
Fri, Nov 13, 2026	Project draft due	Project milestone	11:59 PM
Wed, Nov 18, 2026	Quiz 3 (Related rates)	Quiz	in class
Thu, Nov 19, 2026	Homework 7 (Related rates and linearization)	Assignment due	11:59 PM
Mon, Nov 23, 2026	Test 2 (Derivatives and applications)	Exam	in class
Tue, Nov 24, 2026	Homework 8 (Extrema and MVT)	Assignment due	11:59 PM
Wed, Dec 2, 2026	Quiz 4 (Optimization)	Quiz	in class
Fri, Dec 4, 2026	Final modeling project due	Assignment due	11:59 PM
Wed, Dec 9, 2026	Final exam (cumulative)	Exam	in class

Course policies

Late work and make-ups. Homework is accepted up to 48 hours late for 80% credit; your lowest homework score is dropped. Quizzes and tests cannot be made up without documentation, but your final exam score can replace your lowest test score if it is higher.

Academic integrity. All work you submit must be your own. You may discuss ideas with classmates, but written work, code and solutions must be produced individually unless an assignment says otherwise. Using AI tools to generate submitted work is not allowed unless the assignment explicitly permits it; when it is permitted, you must say how you used it. Suspected violations are reported to the Office of Student Conduct.

Accessibility. Students who need accommodations should contact Disability Resources as early as possible and share their accommodation letter with the instructor. All course materials are available in

accessible formats on request.

Attendance. Regular attendance is expected and strongly linked to success in this course. If you must miss class, review the posted slides and notes and contact a classmate or the instructor. Absences for illness, religious observance or university-sanctioned events are excused with notice.

Communication. Course announcements are posted on the course website. Email the instructor from your university account and include the course code in the subject line; expect a reply within one business day.

Wellness. Your health matters more than any deadline. If you are struggling, reach out early — the Counseling Center, Academic Success Center and your academic advisor are all free resources.

Getting help

Come to office hours early and often — you do not need a specific question. Free campus resources:

- Academic Success Center — free tutoring and study-skills coaching, drop-in or by appointment
- Writing Center — one-on-one help at any stage of a writing project
- Counseling Center — confidential support for stress, anxiety and other concerns
- Library Research Help — chat, email or in-person help finding and citing sources